

Reading the cosmological constant from gas-rich dwarf rotation curves: the a_0 -line and its systematic floor

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ABSTRACT

The radial-acceleration relation of disc galaxies can be written as an exact identity, $E \equiv g_{\text{obs}}^2 - g_{\text{bar}}^2 = a_0 g_{\text{bar}}$: the ‘missing-gravity’ excess is a straight line through the origin whose slope is the low-acceleration scale a_0 at every acceleration, following algebraically from the interpolation kernel $\nu(y) = \sqrt{1 + 1/y}$ of Milgrom (1999). This converts interpolation-function fitting into the measurement of a single slope. We measure that slope on the gas-dominated subsample of the SPARC database (Lelli, McGaugh & Schombert 2016), where the stellar mass-to-light (Υ_*) degeneracy is suppressed by ~ 71 per cent, obtaining $\hat{a}_0 = (0.84\text{--}1.36) \times 10^{-10} \text{ m s}^{-2}$ – the full estimator-and-cut envelope; ± 16 per cent is the 1σ systematic budget about the central value. Inverting through $\Lambda = 32\pi \hat{a}_0^2/c^4$ recovers the *Planck* cosmological constant to within a factor of ~ 2 (central estimates $1.08\text{--}2.03 \times \textit{Planck}$) from galactic kinematics that carry no cosmological input – a sharpened, inversion-form statement of the known $a_0 \sim c\sqrt{\Lambda}$ coincidence. We map the dominant systematics with reproducible pipelines: clean tip-of-the-red-giant-branch distances collapse the estimator-choice systematic but are not the binding wall, and external Υ_* priors tighten a_0 by 2–12 per cent yet leave the measurement systematics-limited even in the perfect- Υ_* limit. The binding floor is gas-mass calibration, which does not average down: the measured box cannot separate the two candidate normalisations of the scale ($cH_\Lambda/Z = 9.36 \times 10^{-11}$ versus $cH_0/Z = 1.13 \times 10^{-10} \text{ m s}^{-2}$, $Z = \sqrt{32\pi/3}$) at 2σ . The order-of-magnitude inversion is robust; the ~ 21 per cent normalisation distinction is data-blocked. The value of a_0 and the sign of the correction are posited inputs throughout, not derived.

Key words: gravitation – dark matter – dark energy – galaxies: kinematics and dynamics – galaxies: dwarf – methods: data analysis

1 INTRODUCTION

1.1 The framework, the horizon scale, and the two footings

Modified inertia (MI) proposes that the inertial response of a body to a force is altered at accelerations below a critical scale a_0 , rather than the gravitational field being modified (Milgrom 1983, 1999). In the de Sitter–Unruh realization considered here, an accelerated observer in a universe with a cosmological horizon experiences a floor in the vacuum’s response set by the horizon temperature, and the transition scale is fixed to the horizon:

$$a_0 = \frac{cH_\Lambda}{Z} = \frac{c^2 \sqrt{\Lambda/3}}{Z}, \quad Z = \sqrt{\frac{32\pi}{3}} = 5.78881, \quad (1)$$

where $H_\Lambda \equiv c\sqrt{\Lambda/3}$ is the de Sitter (pure dark-energy) Hubble rate. With the *Planck* 2018 value $\Lambda = 1.089 \times 10^{-52} \text{ m}^{-2}$ (Planck Collaboration 2020) this gives the **canonical footing**

$$a_0^{\text{canon}} = c^2 \sqrt{\Lambda/32\pi} = 9.355 \times 10^{-11} \text{ m s}^{-2} \quad (2)$$

(*Planck*-anchored, ± 1 per cent). A second, distinct **ALT footing** replaces the pure- Λ density by the total present-day critical density, $a_0^{\text{ALT}} = cH_0/Z = 1.1305 \times 10^{-10} \text{ m s}^{-2}$. A *footing*, in the sense used throughout this paper, is a choice of which cosmological density the horizon term tracks – pure dark energy (ρ_{DE} , giving cH_Λ) versus the total present-day density (giving cH_0). It is a physics choice about the framework, not a fitting freedom. The two footings differ by 20.9 per cent; deciding which one nature uses is one of the two questions this paper confronts, and **both footings are carried on every dimensional number below**. Nothing that follows derives the value of a_0 or the sign of the inertial correction $s = -1$: those remain postulates. What the framework ties to the cosmological horizon is the *scale* of a_0 , and it is that tie – the $\Lambda \leftrightarrow a_0$ relation – that is under measurement.

The proximity of a_0 to a cosmological acceleration is not new: the numerical coincidence $a_0 \sim cH_0$ is Milgrom’s own observation (Milgrom 1983), and the specific proportionality of a_0 to the dark-energy density, $a_0 \propto \sqrt{\rho_{\text{DE}}}$, together with its redshift test, is due to Limbach, Psaltis & Özel (2008). The relativistic completion of the MOND scale as a covariant field theory is the domain of AeST (Skordis & Złośnik 2021), which is modified *gravity*, not the modified-inertia reading adopted

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here. This paper claims none of those; its contribution is the measurement framing developed below.

1.2 What is established, and the question this paper answers

The *forward* statement of the tie – given Λ , predict a_0 – was reported in *The Zimmerman Theory of Gravity* and formalized in the MI field-theory results (Zimmerman 2026a), with the lensing sector constrained separately (Zimmerman 2026b). Independently, the standard MOND phenomenology on the SPARC rotation-curve database is well established: the radial-acceleration relation (RAR; McGaugh, Lelli & Schombert 2016) exhibits a characteristic scale $g^\dagger = 1.2 \times 10^{-10} \text{ m s}^{-2}$, and the SPARC sample itself (Lelli, McGaugh & Schombert 2016) provides the baryonic decompositions used here.

This paper asks the **inverse** question. If a_0 is genuinely a horizon-scale quantity, then galaxy rotation curves – sensitive to a_0 in their low-acceleration outskirts – should let one *read the cosmological constant back out*, via $\Lambda = 3Z^2 a_0^2 / c^4$. Can they? And can the measured a_0 tell the two footings apart? The answer developed below is the crispest single statement of the result: **the order-of-magnitude inversion is real and striking (rotation curves recover Λ to a factor ~ 2 with no cosmological input on the galactic side), but the 21 per cent footing distinction is blocked by an irreducible systematic floor**. We report this as a *measurement* – striking but non-decisive – and we are equally careful not to manufacture a canonical detection nor a deficit: the best current a_0 estimate is a box that straddles *both* footings.

1.3 Credit for the kernel

The interpolation function underlying the identity of Section 2,

$$\nu(y) = \sqrt{1 + 1/y}, \quad y \equiv g_{\text{bar}}/a_0, \quad (3)$$

is **not new to this framework**. It is exactly the kernel written by Milgrom (1999, Eq. 9), in the same functional form, in the original MOND context of Milgrom (1983). Milgrom’s coefficient was $2cH_\Lambda$ rather than cH_Λ/Z . The identity of Section 2 follows from *squaring* that kernel and is therefore a re-expression of a known law; what the present framework contributes, and what this paper measures, is (a) the exact-constancy fingerprint $\varepsilon = 1$, (b) the specific cH_Λ/Z coefficient, and (c) the Λ -inversion. We credit the kernel at the point of use and claim only the distinctive content.

2 THE a_0 -LINE IDENTITY

2.1 The exact identity

The framework’s interpolation (Milgrom’s kernel, Section 1.3) $g_{\text{obs}} = \sqrt{g_{\text{bar}}^2 + a_0 g_{\text{bar}}}$, i.e. $g_{\text{obs}} = g_{\text{bar}} \nu(y)$ with $\nu = \sqrt{1 + 1/y}$, squares to

$$E \equiv g_{\text{obs}}^2 - g_{\text{bar}}^2 = a_0 g_{\text{bar}} \quad \text{exactly, at every } y. \quad (4)$$

Table 1. The excess ratio $\varepsilon(y) \equiv E/(a_0 g_{\text{bar}})$ for the framework kernel and two rivals.

$\nu(y)$	$\varepsilon(y)$	$y \rightarrow 0$	$y \rightarrow \infty$
framework $\sqrt{1 + 1/y}$	1 (exact constant)	1	1
RAR-fit $[1 - e^{-\sqrt{y}}]^{-1}$	$y(\nu^2 - 1)$	1	$\sim 2y e^{-\sqrt{y}} \rightarrow 0$
simple $\frac{1}{2} + \sqrt{\frac{1}{4} + 1/y}$	$1 + \frac{y}{2}(\sqrt{1 + 4/y} - 1)$	1	2

The excess E – the squared difference between observed and baryonic gravity, i.e. the ‘missing gravity’ in acceleration-squared units – is a **straight line through the origin with slope a_0** , valid at all accelerations. There is no deep-MOND selection, no interpolation fit, no small- or large- y approximation: the relation is algebraic and exact (verified symbolically in `identity_uniqueness.py`, Appendix A). This converts the usual task – *fit an interpolation function* – into a simpler one: *measure one slope*.

2.2 Uniqueness, honestly characterized

Demanding an exactly linear excess, $y^2(\nu^2 - 1) = \alpha y + \beta$ for all $y > 0$, gives $\nu^2 = 1 + \alpha/y + \beta/y^2$. The β term produces a non-zero floor, $g_{\text{obs}}^2 \rightarrow \beta a_0^2$ as $g_{\text{bar}} \rightarrow 0$, so requiring $g_{\text{obs}} \rightarrow 0$ forces $\beta = 0$; the surviving family $\nu = \sqrt{1 + \alpha/y}$ is pure a_0 -rescaling, and the standard deep-MOND normalization $g_{\text{obs}} \rightarrow \sqrt{a_0 g_{\text{bar}}}$ fixes $\alpha = 1$. So $\nu = \sqrt{1 + 1/y}$ is the **unique interpolation with an exactly linear excess**, up to the definition of a_0 itself (verified in `identity_uniqueness.py`; re-derived one generalization further in `verify_independent.py`). We state this honestly: the claim is definitionally tight – ‘excess exactly linear through the origin’ is the framework law restated. Its value is methodological, not a new physical fact by itself.

2.3 The $\varepsilon = 1$ fingerprint versus rival kernels: a future-data lever

Define the dimensionless excess ratio $\varepsilon(y) \equiv E/(a_0 g_{\text{bar}}) = y(\nu^2 - 1)$. For the framework, $\varepsilon \equiv 1$ exactly at all y . Rival kernels do not share this (Table 1).

The three agree in the deep regime ($\varepsilon \rightarrow 1$ as $y \rightarrow 0$) but diverge at high y . The McGaugh, Lelli & Schombert (2016) RAR-fit excess **dies superexponentially** – a death that survives any choice of $g^\dagger$. The simple kernel’s excess **persists but at slope $2a_0$** , so $\varepsilon \rightarrow 2$: tail persistence alone does *not* separate the framework from the simple kernel; only the *exact constancy* $\varepsilon \equiv 1$ at *all* y does. This is the framework’s genuine fingerprint. The separation from the RAR fit reaches a factor ~ 110 at $y = 100$ on a matched scale (a factor ~ 11 – 44 at the rival’s own preferred scale – the tail *death* is scale-robust, the exact multiplier is convention-cosmetic).

Two points on the logic. First, the **constancy** of ε across g_{bar} is a scale-independent statement (it needs no value of a_0); the further claim that the constant *equals* a_0 does need a_0 . It is the constancy, not the value, that is the clean fingerprint. Second, the catch, quantified in Section 3, is that SPARC samples this fingerprint thinly: the median point sits at $y = 0.31$, and only one point in the entire quality-cut sample exceeds $y = 100$. The constancy test is in-principle orthogonal to the M/L degeneracy but, at SPARC’s sampling, deliv-

ers less than 1σ of persistent-versus-dying discrimination today (a global framework-versus-RAR-fit shape comparison is a wash, $|\Delta\chi^2| \approx 2$, with M/L profiled; `fire_linearity.py`). **It is a future-data lever, not a present discriminator**, and it is not tested by the data used here.

3 THE GAS-DOMINATED MEASUREMENT

3.1 The gas cut and why it matters

The slope a_0 is estimated by iterated generalized least squares (GLS) through the origin with model-based errors,

$$\hat{a}_0 = \frac{\sum_i w_i E_i g_i}{\sum_i w_i g_i^2}, \quad \text{Var}_{\text{stat}} = \frac{1}{\sum_i w_i g_i^2}, \quad (5)$$

on the SPARC point sample after quality cuts ($Q \leq 2$, inclination $\geq 30^\circ$, $e_V/V < 10$ per cent; frozen read-only database, 175 galaxies). The controlling systematic on the full sample is the stellar mass-to-light ratio Υ : swinging Υ_d over its full physical range $0.5 \rightarrow 0.8$ moves the full-sample $\hat{a}_0$ from $1.89 \rightarrow 1.10 \times 10^{-10}$, a 62 per cent swing that reproduces the known RAR a_0 - Υ degeneracy essentially in full. On the full sample the a_0 -line is an *exact reframing* of the RAR, but it inherits that degeneracy completely and adds no information.

The genuinely new content comes from the **gas-dominated subsample** – the points where the atomic-gas mass (measured from HI, independent of any stellar M/L) dominates the baryonic gravity, so the dominant M/L nuisance is largely removed. Applying a point-level cut

$$V_{\text{gas}}^2 > \Upsilon_d V_{\text{disk}}^2 + \Upsilon_b V_{\text{bul}}^2, \quad \Upsilon_b = 1.4 \Upsilon_d, \quad (6)$$

isolates those points. At the fiducial $\Upsilon_d = 0.7$ this retains 310 points in 49 galaxies (weighted stellar share $\langle \varphi \rangle = 0.32$). On this subsample the absolute Υ swing in $\hat{a}_0$ collapses from 7.90×10^{-11} (full sample) to 2.29×10^{-11} (gas-dominated): **the gas cut removes ~ 71 per cent of the absolute a_0 - Υ degeneracy** (`fire_slope.py`; equivalently, in fractional terms the swing drops from 62 to 19 per cent, a ~ 69 per cent reduction; independently reproduced at 15–29 per cent residual swing in `verify_independent.py`). This is the paper’s single new measurement lever, and it is standard in spirit – gas-rich galaxies as M/L-insensitive probes date to [McGaugh \(2011\)](#); ‘new’ here means new relative to the framework’s own previously established walls.

3.2 The estimate and its honest box

The gas-dominated slope is

$$\hat{a}_0^{\text{gas}} = 1.181 \times 10^{-10} \text{ (GLS)} / 0.973 \times 10^{-10} \text{ (median)}, \quad (7)$$

with independent-machinery estimators (own parser, own cuts, own estimators) landing at $0.92\text{--}0.97 \times 10^{-10}$ – i.e. siding with the median variant. The GLS–median spread is the single largest line in the error budget and is a physics-free degree of freedom; it is charged as a systematic, not hidden. Combining all estimator and cut freedom gives an **honest all-choices box**

$$\hat{a}_0^{\text{gas}} \in (0.84\text{--}1.36) \times 10^{-10} \text{ m s}^{-2}. \quad (8)$$

Two distinct widths are quoted throughout, and they are not the same number: the $(0.84\text{--}1.36) \times 10^{-10}$ box is the

full estimator-and-cut envelope, while ± 16 per cent is the 1σ *systematic budget about the GLS central* (Section 4.3); the envelope is wider than the 1σ budget by construction.

Placed against the two footings at the gas slope’s honest error, the GLS point estimate sits $+1.29\sigma$ above canonical and $+0.27\sigma$ above ALT; the median variant sits $+0.19\sigma$ above canonical and -0.83σ below ALT. Because the independent estimators lean toward the median – the *better-supported* central – and it lands *closer* to canonical, **there is no usable footing lean in either direction**: the 21 per cent fork is not decided. The box straddles both footings.

One estimator bias runs in a known direction. The response $E = g_{\text{obs}}^2 - g_{\text{bar}}^2$ carries an uncorrected positive bias: $\mathbb{E}[g_{\text{obs}}^2] = g_{\text{obs,true}}^2 + \text{Var}(g_{\text{obs}}) > g_{\text{obs,true}}^2$, so $\mathbb{E}[\hat{a}_0] \approx a_0 + \sum w g \text{Var}(g_{\text{obs}}) / \sum w g^2$ – a positive push of order (fractional velocity error) $^2 \approx 1\text{--}4$ per cent. (The committed `gls(biased=False)` weighting cures the *weight*–noise correlation, $\mathbb{E}[w\epsilon] < 0$; it does not debias the *response*.) This residual is small and sub-dominant to the effect of Section 3.3, but it pushes the central in exactly the ‘sits above both anchors’ direction on which the mild disfavouring of canonical rests. Its sign therefore means the canonical footing is, if anything, treated *conservatively* – the true a_0 is slightly lower than the reported central, i.e. slightly closer to canonical, not further from it.

3.3 The declining per-point a_0 : an unmodelled g_{bar} -correlated systematic

A load-bearing caveat, exposed under adversarial verification: the per-point quantity $a_{0,\text{pt}} = E/g_{\text{bar}}$ **declines across the deep regime**. Binning the gas points by g_{bar} (TRGB set, $\Upsilon_d = 0.7$) gives per-point a_0 of $\sim 1.62 \rightarrow 1.24 \rightarrow 0.62 \times 10^{-10}$ from the deepest to the highest g_{bar} tercile – a factor ~ 2.6 decline, entirely within $y \in [0.009, 0.17]$.

It is important to name this correctly, because it is easy to mislabel. The framework’s own fingerprint (Section 2.3) is $\varepsilon \equiv 1$ *exactly at all y* , which is precisely the statement $E/g_{\text{bar}} = \text{const} = a_0$ with **zero shape to leak**: under the framework there is nothing about the kernel that could make $a_{0,\text{pt}}$ decline. Nor do the rivals explain it: at the sampled deep $y \in [0.009, 0.17]$, the RAR-fit and simple kernels both predict ε *rising* with g_{bar} ($\approx 1.13 \rightarrow 1.32$ and $\approx 1.13 \rightarrow 1.34$, respectively) – the *opposite sign* to the observed decline. The decline therefore matches *no* candidate kernel in sign or magnitude. It is not kernel shape; it is an **unmodelled g_{bar} -correlated systematic**, most plausibly residual stellar M/L acting through the φ - g_{bar} correlation at the fixed fiducial $\Upsilon_d = 0.7$ (raising or lowering Υ tilts the trend – a testable signature of M/L, not of the interpolation). Two consequences follow. First, a genuinely declining $a_{0,\text{pt}}$ is a *mild violation* of the framework’s own $\varepsilon \equiv 1$ fingerprint, not a benign feature of it; we book it as a systematic within the estimator-choice line s_{Est} , not as a measured $\varepsilon \neq 1$ (which the data are too shallow in y to establish). Second, because the cause is a fixable estimator/M-L bias rather than irreducible kernel curvature, the estimator-weighted central is a *regime-weighted slope* rather than a clean single a_0 : median/GLS weight the deep regime ($\rightarrow 1.3 \times 10^{-10}$), while a weight-free g_{bar} -OLS weights the high- g_{bar} tail ($\rightarrow 0.89 \times 10^{-10}$, near canonical – though that weighting is wrong for the strongly heteroscedastic E). A future E -debiased or forward-modelled estimator,

or points reaching $y \sim 1$, could recover a cleaner single a_0 . The estimator-choice systematic $s_{\text{Est}} = |\text{GLS} - \text{median}|/2 \approx 1.04 \times 10^{-11}$ books this spread, and on the full-gas sample it alone exceeds the footing-splitting threshold. The honest consequence is that the measured a_0 is intrinsically a **box straddling both footings**, not a single-footing detection – and the low g_{bar}^2 -OLS value is likewise *not* a canonical detection.

4 THE SYSTEMATIC MAP: DISTANCE AND MASS-TO-LIGHT

Three independent levers could in principle split the footings on current SPARC – the radial-acceleration relation itself, distance quality, and stellar M/L. The first is convention-compatible and non-diagnostic of a_0 's exact value (Section 3.1); the second and third are mapped and spent below (Sections 4.1 and 4.2). All three fail on the same wall, which is why the fork is BIG-SPARC-gated (Section 6). Each lever below carries a dedicated, committed, exit-0 pipeline and an adversarial verification pass. The threshold that matters throughout is the one needed to split the two footings at 2σ :

$$\Delta = |a_0^{\text{ALT}} - a_0^{\text{canon}}| = 1.951 \times 10^{-11} \text{ m s}^{-2}, \quad (9)$$

$$\sigma_{\text{tot}} \leq |\Delta|/2 = 9.753 \times 10^{-12} \text{ m s}^{-2}.$$

4.1 The TRGB distance lever

Distances enter the excess through $g_{\text{obs}} \propto 1/D$ (while, exactly, $g_{\text{bar}} \propto D^0$ for gas *and* stars, since baryonic surface density is distance-independent – verified symbolically). SPARC's distance-method flag f_D isolates galaxies with tip-of-the-red-giant-branch (TRGB) or Cepheid distances ($f_D \in \{2, 3\}$, distance error ~ 5 per cent) from the Hubble-flow majority ($f_D = 1$, ~ 25 per cent). Restricting the gas-dominated subsample to $f_D \in \{2, 3\}$ retains 18–20 galaxies (147–191 points), roughly half the gas sample and point-balanced against the Hubble-flow set – so this is a *budget* test, not a sample-size test (`scout_split.py`). Crucially, the TRGB and Hubble-flow subsamples occupy the *same* acceleration window (y -medians 0.037 versus 0.041): the identity $E = a_0 g_{\text{bar}}$ holds at every g_{bar} regardless of a galaxy's mass or distance, so the two probe the same segment of the a_0 -line.

The lever works mechanically (`est_gls.py`, `est_robust.py`; verified in `verify_trgb.py`):

(i) **The distance systematic is cut** ~ 2 – $3.5 \times (6.5 \rightarrow 2.8$ per cent fractional at $\Upsilon_d = 0.7$).

(ii) **The estimator-choice systematic collapses**: on the clean-distance set the median and GLS estimators agree to $\sim 0.2\sigma$ (median 1.273, GLS 1.333×10^{-10} at $\Upsilon_d = 0.7$).

(iii) **A Hubble-flow artifact is caught**: the banked low full-gas median (0.973×10^{-10}) was largely a Hubble-flow-distance artifact; the clean-distance central moves *up* coherently across median, GLS, and bootstrap to ~ 1.27 – 1.35×10^{-10} (Hubble-flow-only median 0.805×10^{-10}). This is robust (galaxy-level bootstrap 1.329 [16–84 per cent: 1.173, 1.486] $\times 10^{-10}$; maximum leave-one-out shift < 10 per cent) and is a distance-*scale* effect, not a g_{bar} -segment selection artifact (range-matching trims zero points).

But the upward move is itself estimator-weighting-dependent (the unmodelled g_{bar} -correlated systematic of Section 3.3: the weight-free OLS gives 0.89×10^{-10}), so it is a regime-weighted slope of uncertain origin, not a canonical detection nor a manufactured deficit. And, decisively, **distance is not the binding wall**: once gas-dominated, the total error is floored by the M/L and gas-calibration systematics *above* the $|\Delta|/2$ target on every prior convention. The clean-distance central at $\sim 1.3 \times 10^{-10}$ sits above *both* anchors, leaning mildly *against* canonical (-2.75σ) rather than detecting it, with ALT weakly favoured (-1.28σ) but not confirmed – and, per Section 3.3, that very overshoot is partly the unmodelled systematic rather than a clean signal, so it is not read as a real move against canonical. The footing separation is ≤ 1.3 bans (log-flat), < 1 ban under an adversarial informed prior – below the 2-ban line on every convention. **Verdict: the TRGB lever tightens the method but is non-diagnostic of the footing; the distance flag is spent.**

4.2 The mass-to-light lever

The residual Υ uncertainty ($\sigma_{\ln \Upsilon} = 0.0999$ dex) enters the current budget as *one global, fully coherent* number, so it does not average down with galaxy count – it is a floor. The lever (`setup_mlpriors.py`, `est_indep.py`, `est_marg.py`, `est_wall.py`; verified) decomposes it into a **coherent SPS/IMF zero-point floor** (irreducible; external colours cannot touch it) plus a **per-galaxy reducible part** (which external [3.6]+colour SPS priors shrink, and which averages $\sim 1/\sqrt{N}$). Two calibration-preserving literature decompositions (quadrature $\approx$ the banked 0.0999 dex; Schombert, McGaugh & Lelli 2019; Meidt et al. 2014; McGaugh & Schombert 2014; Bell & de Jong 2001) are used, since SPARC ships one fixed Υ per galaxy and no local colour vector:

- balanced: coherent 0.060 dex + per-galaxy residual 0.040 dex;
- NIR-realistic: coherent 0.075 dex + per-galaxy residual 0.035 dex.

Three findings, each verified both ways:

(i) **The real gain is recognizing the split, not the external prior**. Because at [3.6] the M/L is nearly constant, the per-galaxy reducible signal is intrinsically small; the coherent line drops from 9.57×10^{-12} to $\sim 5.8 \times 10^{-12}$ (balanced, $\Upsilon_d = 0.7$) mostly by acknowledging that part of the M/L error already averaged over ~ 49 galaxies, not by new colour data.

(ii) **The coherent Υ zero-point is only partially degenerate with a_0** . Treating it as a proper Bayesian nuisance template ($C = \text{diag}(\sigma_i^2) + s^2 U U^\top$, Sherman–Morrison marginalization) gives correlation $\rho(a_0, \alpha) = -0.65$ to -0.81 , *never* $\rightarrow 1$, because the stellar share φ genuinely varies with g_{bar} so the template U is not proportional to g . The data therefore self-calibrate the nuisance, and marginalization **tightens a_0** by 2–12 per cent (marg/quad = 0.88–0.98): the quadrature budget was mildly conservative, not anti-conservative.

(iii) **Even perfect M/L is insufficient**. Zeroing the Υ systematic *entirely* still leaves the best-case total at $\sim 12.96 \times$

Table 2. The gas-dominated systematic budget at $\Upsilon_d = 0.7$ (all σ in $10^{-12} \text{ m s}^{-2}$). Threshold to split the footings at 2σ : $\sigma_{\text{tot}} \leq 9.753 \times 10^{-12}$. No configuration reaches it.

Systematic line	full-gas (banked)	full-gas + Υ lever	TRGB + Υ lever
statistical	4.67	4.67	6.12
distance (s_D)	7.63	7.63	3.80
inclination (s_i)	2.64	2.64	4.22
stellar M/L (s_U)	9.57	5.79	6.83
gas calibration (s_G)	8.63	8.63	9.46
estimator/regime-slope (s_{Est})	10.44	10.44	3.00
total σ_{tot}	19.03	17.43	14.65
(per cent of $\hat{a}_0$)	16.1	14.8	11.0
footing separation	$\sim 1.2\sigma$	$\sim 1.3\sigma$	$\sim 1.7\sigma$

10^{-12} (TRGB, $\Upsilon_d = 0.7$; $\sim 1.94\sigma$ footing separation) – above 9.753×10^{-12} and under 2σ everywhere. Beating Υ is *necessary but not sufficient*.

The reason is that **the true wall after Υ is gas-mass calibration** (s_G , coherent $\sigma_{\text{in } G} = 0.10$). It does *not* average down and does *not* self-calibrate, because on gas-dominated points $1 - \varphi \approx 1$, so the gas template is nearly proportional to g . The coherent floor, $\text{hypot}(5.75, 8.63) \times 10^{-12}$,

$$\text{hypot}(s_{U,\text{coh}}, s_G) = 1.037 \times 10^{-11} > |\Delta|/2 \quad (10)$$

exceeds the footing-splitting threshold at any N at $\sigma_{\text{in } G} = 0.10$ (coherent terms do not average down). An external swap to colour-based M/L (McGaugh & Schombert 2014) trades the SPS coherent zero-point for a comparable colour-M/L/dust/age-metallicity zero-point (~ 0.10 – 0.15 dex) with no net gain (a swept coherent floor never crosses the threshold). **Verdict: the M/L lever tightens a_0 but is non-diagnostic of the footing; the wall shifts to gas-mass calibration.**

4.3 The systematic budget

Table 2 gives the gas-dominated budget at $\Upsilon_d = 0.7$ (310 points, 49 galaxies; all σ in $10^{-12} \text{ m s}^{-2}$), before and after the two levers. Each column is a single, internally consistent configuration: full-gas (banked coherent Υ), the Υ -lever applied to the full-gas set (balanced decomposition), and the Υ -lever applied to the TRGB clean-distance set (balanced decomposition).

Footing separation here is the log-space metric $|\ln(a_0^{\text{ALT}}/a_0^{\text{canon}})|/(\sigma_{\text{tot}}/\hat{a}_0) = 0.1895/\sigma_{\text{in,tot}}$ – how many log- σ separate the two footings – evaluated at each column’s central $\hat{a}_0$ (full-gas 1.181, TRGB 1.333×10^{-10}). It is the relevant ‘can I split them’ number and is not the same as the linear ratio $|\Delta|/\sigma_{\text{tot}}$, which is ≈ 1.0 for full-gas; both are below 2. Threshold to split at 2σ : $\sigma_{\text{tot}} \leq 9.753 \times 10^{-12}$. **No configuration reaches it** – not the TRGB clean-distance set, not the external- Υ lever, not the two combined, and not even the $s_U \rightarrow 0$ perfect-M/L limit ($\sim 12.96 \times 10^{-12}$, 1.94 σ ; Section 4.2). The binding lines after both levers are gas calibration $s_G \approx 8.6$ – 9.5 , the estimator/regime-slope spread s_{Est} , and the coherent SPS floor – none of which distance or M/L work reduces. (For reference, the banked TRGB set *before* the Υ lever carried $s_U = 11.18$ and

$\sigma_{\text{tot}} = 17.1 \times 10^{-12}$, 12.8 per cent of $\hat{a}_0$; the lever is what moves the TRGB column to 14.65.)

5 THE Λ -INVERSION

The horizon tie inverts algebraically to the cosmological constant:

$$a_0 = \frac{c^2 \sqrt{\Lambda/3}}{Z} \iff \Lambda = \frac{3Z^2 \hat{a}_0^2}{c^4} \quad (11)$$

(sympy-inverted, `fire_lambda.py`; re-derived by hand in verification; note $3Z^2 = 32\pi$, so equivalently $\Lambda = 32\pi \hat{a}_0^2/c^4$).

What is and is not tested by this inversion – the load-bearing point. Because the canonical $a_0^{\text{canon}} \equiv c^2 \sqrt{\Lambda_{\text{Planck}}/32\pi}$ is defined from *Planck*’s Λ , the predicted ratio is *identically*

$$\frac{\Lambda_{\text{pred}}}{\Lambda_{\text{Planck}}} = \left(\frac{\hat{a}_0}{a_0^{\text{canon}}} \right)^2 \quad \text{exactly.} \quad (12)$$

The inversion is therefore a *single-number test* – does the rotation-measured a_0 equal the horizon value cH_Λ/Z ? – squared and re-expressed, not many independent decades of confirmation. The ‘factor ~ 2 in Λ ’ is the ‘factor ~ 1.4 in a_0 ’ squared. What *is* genuinely independent and non-trivial is that the rotation-curve $\hat{a}_0$ uses *zero* cosmological input (only rotation velocities V , radii R , and baryonic decompositions), so its agreement with a horizon-derived acceleration is a real galactic-versus-cosmological coincidence – the classic Milgrom $a_0 \sim cH$ coincidence, here sharpened. What is *not* tested is (a) the normalization $Z = \sqrt{32\pi/3}$, which appears on *both* sides of the inversion and is assumed throughout, never checked against data; and (b) the ‘large exponent’ of Λ ($\sim 10^{-52} \text{ m}^{-2}$ in SI units), which is a unit-dependent dimensional statement, not a measured dynamic range. We flag both so the coincidence is not over-read.

Feeding the gas-dominated slope through the relation gives Table 3.

The clean Λ statement is on the canonical (pure- Λ) footing; the ALT footing ties a_0 to H_0 (matching the GLS slope at $+0.27\sigma$) and does not itself produce a ‘ Λ ’, so the inversion column is canonical-only by construction.

The headline is the coincidence itself: **rotation curves of gas-rich dwarfs recover the cosmological constant to a factor ~ 1.1 – 2.0 of the value measured from the cosmic microwave background – two determinations separated by the span from galactic to cosmological scales, with no cosmological input on the galactic side.** This is what a horizon-scale $a_0 = cH_\Lambda/Z$ predicts, and it is the same information as the previously reported $a_0 \sim cH_\Lambda/Z$ coincidence, reframed as an inversion – now with a sharper falsification target: a future gas-dominated slope at 3×10^{-10} would break it outright. Two honest hedges. First, the *better-supported* estimators (median and independent machinery) land *closer* to *Planck* ($+0.24\sigma$), so the inversion is if anything cleaner than the GLS headline. Second, the *cleaner the distances, the more the central overshoots Planck*: the TRGB-tightened central sits $2.03\times$ high ($+2.76\sigma$), further above rather than onto *Planck*, so the sharpest current data mildly *disfavour* the canonical footing rather than confirm it – and, per Section 3.3, that overshoot is itself partly the

Table 3. The Λ -inversion against *Planck* 2018, $\Lambda = 1.089 \times 10^{-52} \text{ m}^{-2}$. Each row is evaluated at its own configuration’s log-error $\sigma_{\ln \Lambda} = 2\sigma_{\ln a_0}$ from the committed pipeline: 0.32 for the two full-gas rows, 0.26 for the tighter TRGB clean-distance budget (`est_gls_results.json`, $s_{\ln \Lambda} = 0.257$).

Estimate	$\hat{a}_0$ ($10^{-10} \text{ m s}^{-2}$)	Λ_{pred} (10^{-52} m^{-2})	ratio to <i>Planck</i>	$\sigma_{\ln \Lambda}$	tension
full-gas GLS	1.181	1.74	1.59×	0.32	+1.45 σ
full-gas median / independent	0.973	1.18	1.08×	0.32	+0.24 σ
TRGB clean-distance	1.33	2.21	2.03×	0.26	+2.76 σ

unmodelled g_{bar} -correlated systematic, so its interpretation is uncertain and it is not read as a real deficit either.

The Occam comparison. Contrasting the zero-parameter horizon prediction $M_0 = \{a_0 \equiv cH_\Lambda/Z, \text{ no free parameters, } \pm 1 \text{ per cent anchor}\}$ against a free- a_0 model M_1 with a log-flat prior of width W :

$$B_{01} = \underbrace{\frac{W}{\sqrt{2\pi}s}}_{\text{Occam}} \underbrace{e^{-t^2/2}}_{\text{fit penalty}}, \quad t = \frac{\ln a_0^* - \ln \hat{a}_0}{\sqrt{s^2 + s_{\text{anchor}}^2}}, \quad s = 0.161. \quad (13)$$

On the default two-decade prior: $B_{01} = +0.60$ bans (canonical) / +1.04 bans (ALT), with a prior/estimator envelope +0.30 to +1.38 (canonical) / +0.57 to +1.34 (ALT). On Jeffreys’ scale this is ‘substantial’ – **positive but modest, explicitly not decisive**. It formalizes *predicted-not-fitted* (that a_0 was fixed from c, H_Λ, Z before the fit), not new data, and it is prior-convention-fragile: a literature-informed prior conditioned on the MOND scale can drive the canonical bans toward ~ 0 (though never negative). Two-sided, the lever is also a genuine falsification exposure: if a future clean measurement holds the GLS central (1.18×10^{-10}) at $3\times$ smaller error, the canonical footing would move to -2.45 bans – which is precisely what makes the measurement worth sharpening.

6 THE FOOTING FORK AND HONEST SCOPE

The two footings are 20.9 per cent apart:

$$\begin{aligned} a_0^{\text{canon}} &= cH_\Lambda/Z = 9.355 \times 10^{-11} \text{ m s}^{-2} (\rho_{\text{DE}}), \\ a_0^{\text{ALT}} &= cH_0/Z = 1.1305 \times 10^{-10} \text{ m s}^{-2} (\rho_{\text{total}}). \end{aligned} \quad (14)$$

The measured a_0 is a box $(0.84\text{--}1.36) \times 10^{-10}$ that **straddles both footings**. On the estimator-weighted central the canonical footing is mildly *disfavoured* (the central sits above it, $\sim -3\sigma$ high-side across configurations), while ALT is *not confirmed* ($\sim -1.4\sigma$). We manufacture neither a canonical detection nor a deficit: the honest reading is a straddle, and the mild high-side lean is partly the response bias (Section 3.2) and the unmodelled g_{bar} -correlated systematic (Section 3.3), both of which push the central up.

The fork is **not decidable from current SPARC by any systematic lever**. The RAR is convention-compatible and non-diagnostic of a_0 ’s exact value; the TRGB distance lever is spent (Section 4.1); the M/L lever is spent, including its perfect-M/L limit (Section 4.2). The binding wall is the coherent gas-calibration + SPS floor + estimator/regime-slope spread, and $\text{hypot}(s_{U,\text{coh}}, s_G) = 1.037 \times 10^{-11}$ exceeds $|\Delta|/2 = 9.753 \times 10^{-12}$ at *any* N at the current gas-calibration precision.

Deciding the fork therefore requires *all* of:

(i) **BIG-SPARC-scale statistics** – a clean-distance gas-dwarf sample of $N \sim 300\text{--}670$ (the what-it-takes map: at $\sigma_{\ln G} = 0.08$, $N \sim 666$; at 0.06, $N \sim 273$), to pull the statistical floor down and force the estimator-choice systematic into the open;

(ii) **an independent gas-mass calibration** cutting $\sigma_{\ln G}$ from 0.10 to $\lesssim 0.08$ (interferometric H I + CO, He/metal corrections) – since at $\sigma_{\ln G} = 0.10$ no finite N suffices;

(iii) **points reaching $y \sim 1$** to break the regime-slope magnitude systematic of Section 3.3 (and, as a bonus, to make the $\varepsilon = 1$ constancy fingerprint of Section 2.3 testable).

What is robust today is the **order-of-magnitude inversion**: Λ recovered from dwarf rotation to within a factor ~ 2 of *Planck*, with no cosmological input on the galactic side. What is blocked is the **21 per cent which-dark-energy-quantity distinction** – canonical ρ_{DE} versus total ρ_{total} . The first is a striking measurement; the second is data-gated, and we state it as such.

A footing-fork subtlety worth flagging for future epochs: the canonical footing predicts the gas slope tracks $\sqrt{\rho_\Lambda}$ (constant in redshift), while the ALT footing tracks $cH(z)E(z)$ (rising). A multi-epoch $a_0(z)$ measurement (e.g. via DESI-scale samples) would turn the 21 per cent degeneracy into a slope-versus-redshift discriminator – the one route that does not run into the static systematic floor mapped here.

7 DISCUSSION

The result of this paper is a **measurement**, and it should be read as one. The a_0 -line identity converts the framework’s rotation-curve law into a single measurable slope; the gas-dominated cut on SPARC removes ~ 71 per cent of the stellar M/L degeneracy that otherwise dominates such measurements; and inverting the slope recovers the cosmological constant to a factor ~ 2 of the *Planck* value, from galactic-scale kinematics that carry no cosmological input. That coincidence – that the outskirts of nearby gas-rich dwarfs encode, to within a factor of two, the same Λ that governs the accelerating expansion of the universe – is the striking positive of the framework’s horizon-scale $a_0 = cH_\Lambda/Z$ hypothesis, and it is stated here without overclaim. Its honest weight is exactly that of the classic Milgrom $a_0 \sim cH$ coincidence and of the $a_0 \propto \sqrt{\rho_{\text{DE}}}$ proposal of Limbach, Psaltis & Özel (2008), sharpened into an inversion and a falsifiable target – no more, because the inversion is one comparison squared, and it assumes rather than tests the normalization Z .

What the measurement does *not* do is equally important. It does not distinguish the two dark-energy footings: the measured a_0 is a box straddling both, and every systematic lever available on current SPARC (the RAR, distance, M/L) has

been mapped and spent without crossing the 2σ splitting threshold. The binding wall is gas-mass calibration together with an irreducible coherent SPS floor and an unmodelled g_{bar} -correlated systematic that pulls the inferred magnitude down across the deep regime – a mild tension with the framework’s own $\varepsilon \equiv 1$ fingerprint, not a benign feature of it. None of these is reducible by more distance or M/L work, and even a perfect M/L measurement leaves the footing separation below 2σ . The identity is a re-expression of a kernel due to Milgrom (1999); the distinctive empirical content is the exact-constancy fingerprint (data-starved today), the cH_Λ/Z coefficient, and the inversion. And the value of a_0 and the sign $s = -1$ are postulates: what is tied to the horizon is the scale.

Read as a statement about method, ‘ a_0 -from-rotation = the dark-energy density scale’ is the sharpest available single-number route from galaxy rotation to the cosmological constant, and this paper strengthens the *method* (the gas cut, the distance and M/L maps, the inversion) while being explicit that it does not yet strengthen the *claim* to a footing-level detection. A genuinely striking numerical coincidence has been sharpened into a falsifiable target, and the two-sided Occam lever means the framework now carries real falsification exposure on the canonical footing – which is what makes the next generation of data (BIG-SPARC, an independent gas-mass calibration, and $y \sim 1$ points) worth acquiring. An in-principle modified-gravity-impossible discriminator between the modified-inertia and modified-gravity readings – the relational velocity-dispersion spread – is developed separately (Zimmerman 2026c), but is not yet detectable with existing data.

8 CONCLUSIONS

Central claim. Gas-rich dwarf-galaxy rotation curves invert, through the fixed constant $Z = \sqrt{32\pi/3}$, to the cosmological constant measured by *Planck* to within a factor of ~ 2 – a striking coincidence with no cosmological input on the galactic side – but the measured $a_0 = (0.84\text{--}1.36) \times 10^{-10} \text{ m s}^{-2}$ is a systematics-limited box that straddles both the canonical ($cH_\Lambda/Z = 9.36 \times 10^{-11}$) and ALT ($cH_0/Z = 1.13 \times 10^{-10}$) footings; the ~ 21 per cent distinction between which dark-energy quantity the horizon term tracks is blocked by a gas-mass-calibration floor and cannot be decided on current SPARC.

(i) **The identity.** Squaring the framework’s interpolation gives $E \equiv g_{\text{obs}}^2 - g_{\text{bar}}^2 = a_0 g_{\text{bar}}$, exact at every acceleration – the missing-gravity excess is a straight line through the origin with slope a_0 . The kernel is Milgrom’s (1999); the distinctive content is the exact constancy $\varepsilon = 1$, the cH_Λ/Z coefficient, and the inversion.

(ii) **The measurement.** On the gas-dominated SPARC subsample the slope is $\hat{a}_0 = (0.84\text{--}1.36) \times 10^{-10} \text{ m s}^{-2}$ (the full estimator-and-cut envelope; ± 16 per cent 1σ systematic budget; no usable footing lean), with ~ 71 per cent of the absolute M/L degeneracy removed and an unmodelled g_{bar} -correlated systematic (a mild tension with $\varepsilon \equiv 1$) carried throughout.

(iii) **The systematic floor.** The TRGB distance lever collapses the estimator-choice systematic and catches a Hubble-flow artifact but is not the binding wall; the M/L

lever tightens a_0 by 2–12 per cent through partial self-calibration ($\rho \approx -0.7$ to -0.8) yet is insufficient even in the perfect-M/L limit. The true wall is gas-mass calibration: $\text{hypot}(s_{U,\text{coh}}, s_G) = 1.037 \times 10^{-11}$ exceeds $|\Delta|/2 = 9.753 \times 10^{-12}$ at any N .

(iv) **The inversion.** $\Lambda = 3Z^2\hat{a}_0^2/c^4$ recovers the cosmological constant to a factor ~ 2 of *Planck* (full-gas GLS $1.59\times$, $+1.45\sigma$; median/independent $1.08\times$, $+0.24\sigma$; TRGB $2.03\times$, $+2.76\sigma$ – overshooting) – the striking positive, favoured by $+0.60$ bans (canonical) / $+1.04$ bans (ALT) on a log-flat Occam comparison, substantial but not decisive. It is one comparison, $\hat{a}_0$ versus cH_Λ/Z , squared and re-expressed, and it assumes the normalization Z .

(v) **The scope.** The order-of-magnitude inversion is robust; the 21 per cent canonical-versus-ALT footing distinction is data-blocked, requiring BIG-SPARC-scale clean-distance samples, a sharper gas-mass calibration ($\sigma_{\ln G} \lesssim 0.08$), and $y \sim 1$ points. The value of a_0 and the sign $s = -1$ remain postulates; what is tied to the horizon is the scale.

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This work made use of the SPARC database (Lelli, McGaugh & Schombert 2016); we thank its authors for making the rotation-curve decompositions publicly available.

DATA AND SOFTWARE AVAILABILITY

This work is based entirely on the public SPARC database (Lelli, McGaugh & Schombert 2016), available at <http://astroweb.cwru.edu/SPARC/>. All load-bearing numbers in this paper reproduce from committed, exit-0 Python scripts in the public repository <https://github.com/carlzimmerman/zimmerman-formula> (directories `prep_2026/a0_line`, `prep_2026/a0_line_trgb`, and `prep_2026/a0_line_mlpriors`); the repository is self-verifying, and Appendix A maps each number to its script. The a_0 -line identity and slope estimator are additionally implemented in the `a0kit` package, archived at Zenodo (DOI 10.5281/zenodo.21478981; ASCL submission under editor review). The manuscript is archived as a preprint at Zenodo (DOI 10.5281/zenodo.21419735).

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APPENDIX A: COMMITTED, EXIT-0 VERIFICATION SCRIPTS

Every load-bearing number above reproduces from raw SPARC data (frozen, read-only) via a committed Python script in the public repository (see Data and Software Availability); each exits 0, carries both footings, and was re-derived by an independent verifier with its own parser, cuts, and estimators. No hard-coded checks: every assertion is a live sympy simplification or a computed numeric comparison. The a_0 -line identity and slope are additionally implemented in the **a0kit** package (Zenodo DOI 10.5281/zenodo.21478981).

The a_0 -line core (prep_2026/a0_line/).

- **identity_uniqueness.py** – sympy verification of the identity $E = a_0 g_{\text{bar}}$, the uniqueness family $\nu = \sqrt{1 + \alpha/y}$ with $\alpha = 1$ forced, the rival excess formulas and $\varepsilon(y)$ table, and the SPARC y -sampling census (median $y = 0.31$, $N(y > 100) = 1$).

- **estimator_theory.py** – the GLS estimator, the observed-error weighting trap ($\mathbb{E}[w\epsilon] < 0$, a $\times 3$ artifact, cured by model-based weights), the residual positive response bias (1–4 per cent, Section 3.2), the exact distance scaling ($g_{\text{bar}} \propto D^0$ for gas and stars), the per-point sensitivity arms, and the full budget.

- **bayes_setup.py** – the Occam factor closed form and quadrature, prior sensitivity, footing comparison, and the Λ -inversion.

- **fire_slope.py** – the gas-dominated GLS slope (1.181×10^{-10} / median 0.973), the 71 per cent absolute degeneracy kill, both footings.

- **fire_linearity.py** – the $\varepsilon = 1$ fingerprint versus the RAR-fit and simple kernels, the shape-test wash ($|\Delta\chi^2| \approx 2$ with M/L profiled).

- **fire_lambda.py** – $\Lambda = 3Z^2 \hat{a}_0^2 / c^4$, 1.74×10^{-52} (GLS) / 1.18×10^{-52} (median) versus *Planck*.

- **fire_occam.py** – $B_{01} = +0.60$ / $+1.04$ bans, the envelope, the two-sided error-reduction lever.

- **fire_common.py**, **verify_independent.py** – shared machinery and the independent-verifier re-derivation.

The TRGB distance lever (prep_2026/a0_line_trgb/).

- **scout_split.py** – the f_D split (18–20 TRGB/Cepheid gas galaxies), same-acceleration-window check.

- **est_gls.py**, **est_robust.py** – the clean-distance slope (1.33×10^{-10} , estimator collapse to 0.2σ , Hubble-flow artifact), both footings.

- **est_forecast.py** – the realized error-reduction / Occam-ban forecast.

- **verify_trgb.py** – independent re-derivation; the g_{bar}^2 -OLS catch of the declining per-point a_0 (deep-to-high tercile $1.62 \rightarrow 0.62 \times 10^{-10}$).

The M/L lever (prep_2026/a0_line_ml priors/).

- **setup_ml priors.py** – the coherent/per-galaxy Υ decomposition, the coherence check (fully-coherent 9.57 versus fully-independent 1.66×10^{-12}).

- **est_indep.py**, **est_wall.py** – the residual-floor and what-it-takes maps; the $s_U \rightarrow 0$ hard limit ($\sim 1.94\sigma$, insufficient); gas-calibration as the binding wall.

- **est_marg.py** – the Bayesian nuisance marginalization ($\rho = -0.65$ to -0.81 ; marginalization tightens 2–12 per cent).

All three lanes returned the verdict **upheld** under adversarial verification, with the honesty rails run in both directions: three artifacts (an observed-weight $\times 3$ fake deficit, a cross- Υ covariance fake $\Delta\chi^2$, and a loose error-reduction ‘win’ forecast) were caught in-script and corrected rather than relayed. No configuration manufactures a footing detection or a deficit; the measured a_0 is a box straddling both footings.

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